

Relationships between variables



1

- a Yes b No c No d Yes

2

- a No.
b Distance from the equator affects temperature.
c No.
d Distance covered affects time of travel or time of travel affects distance covered.
e No
f Time spent working affects wages earned.
g Number of pages affects time spent to finish a novel.
h No
i No.
j No
k Number of pets affects time spent caring for pets (hours).
l Time spent training (hours) affects athletic performance.

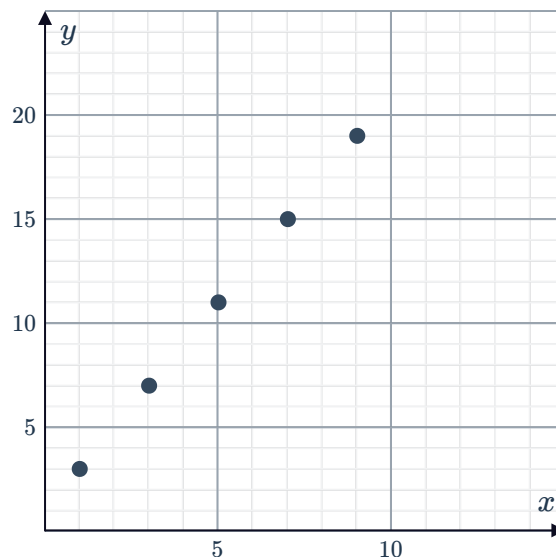
3 Negative

Scatter plots

4

- a Yes b No c No

5



6

- a Strong positive correlation
- c Strong negative correlation
- e Weak negative correlation



- b No correlation
- d Weak positive correlation

7 There is no direct relationship

8

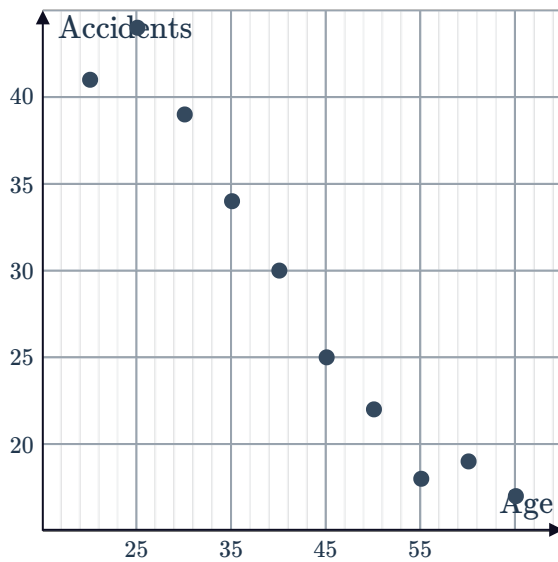
- a As air temperature increases, sea temperature increases.
- b Strong, positive linear correlation

9

- a Amount of healthy coral
- b Sea temperature

10

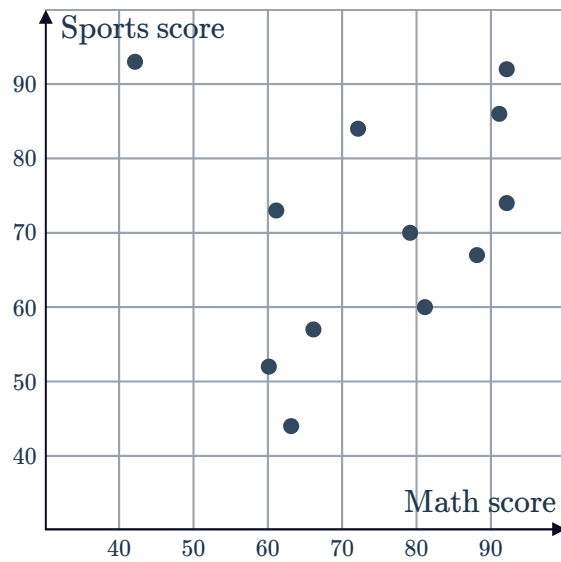
- a
- b Strong, negative, linear



- c No

11

a



b

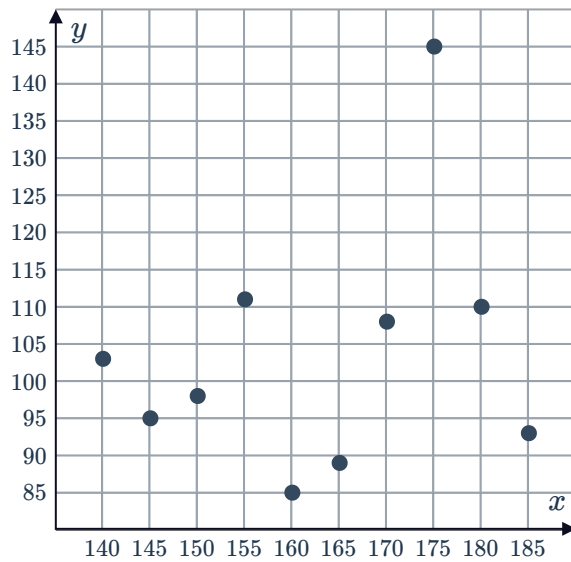
Weak, positive, linear

c

Student 10's

12

a



b

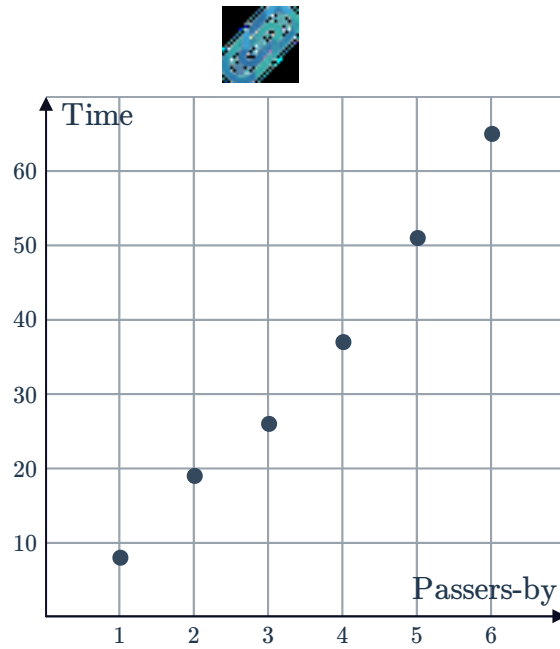
Not correlated

c

Height = 175 cm

13

a

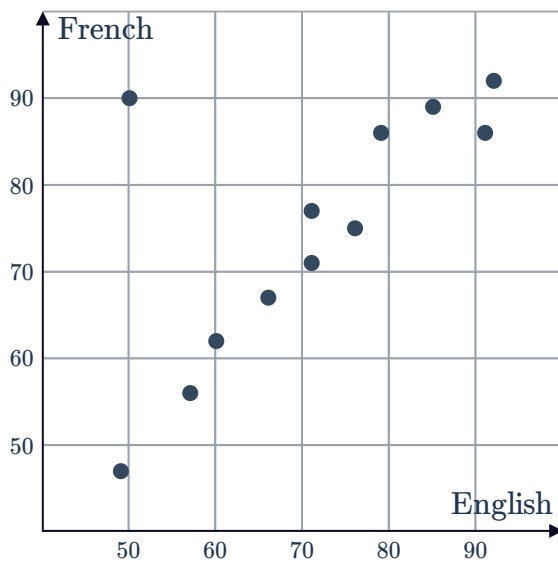


b The more passers-by there are, the longer will be the time until help is offered.

c Strong, positive, linear

14

a

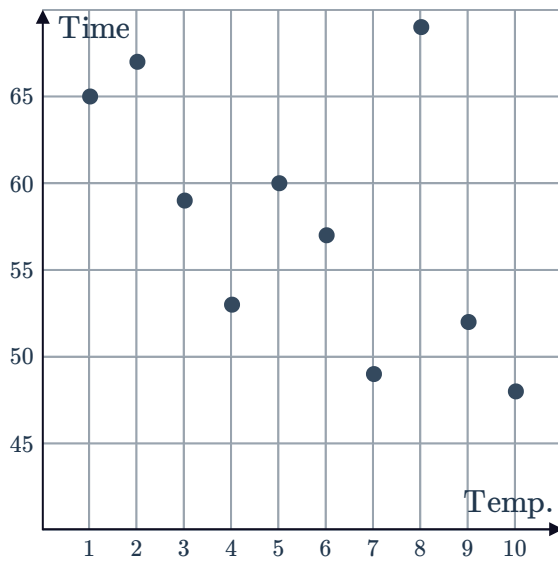


b Strong, positive, linear

c Student 9's

15

a



b 10

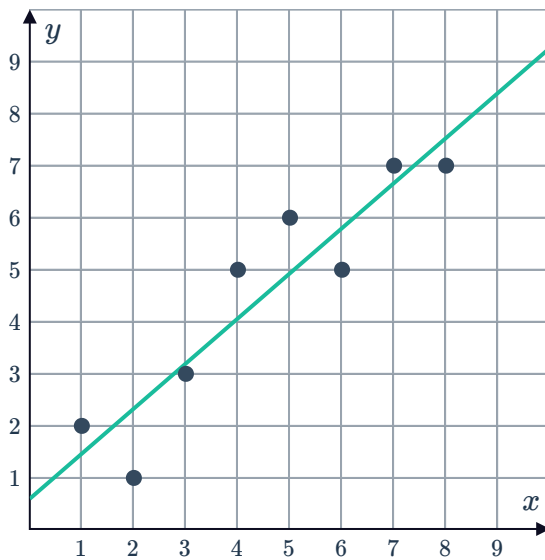
c Moderate, negative, linear

d 8°C , 69 seconds

Lines of best fit

16

a



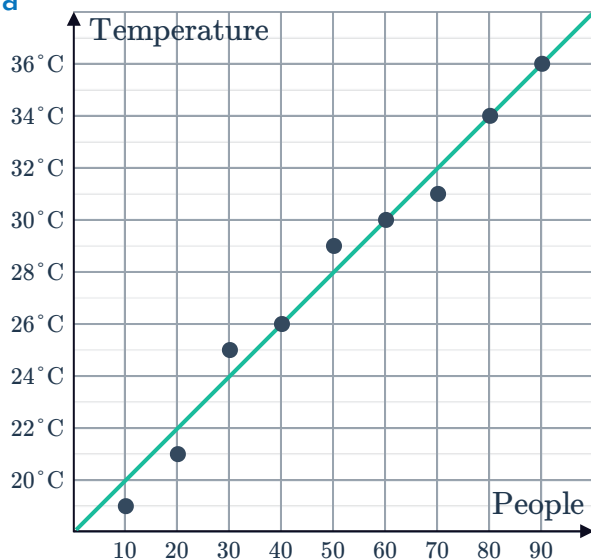
b

i Approx. 4.5

ii Approx. 8.4

17

a



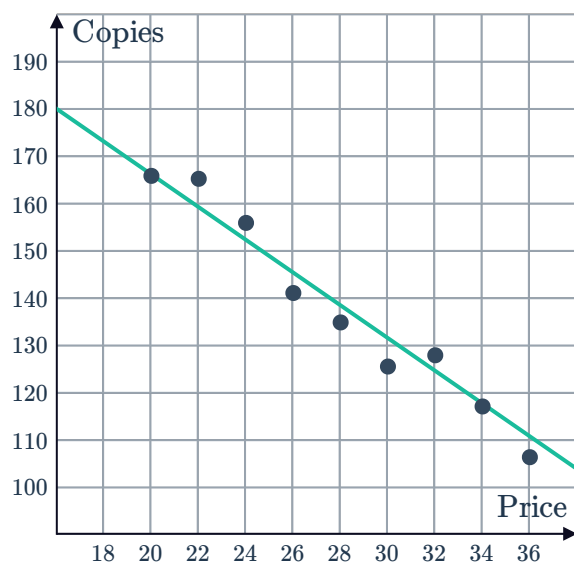
b

- i Approx. 25°C
- ii Approx. 38°C



18

a

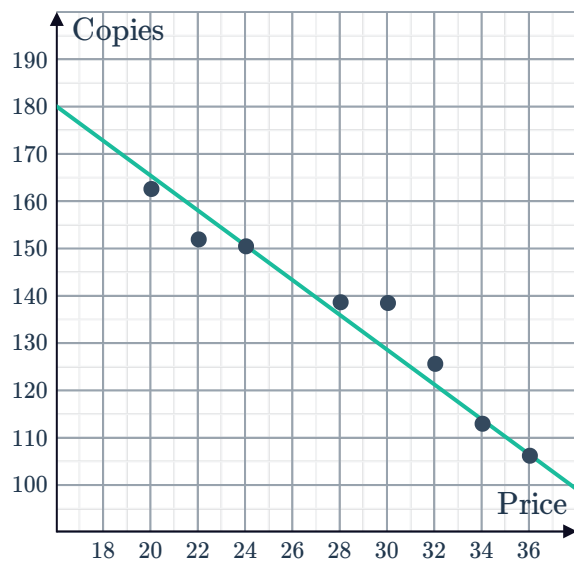


b

- i Approx. 121
- ii Approx. 174

19

a



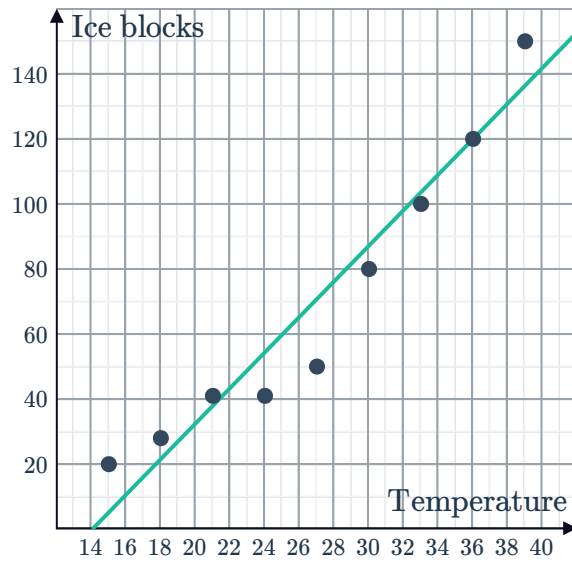
b

- i Approx. 117
- ii Approx. 173



20

a



b

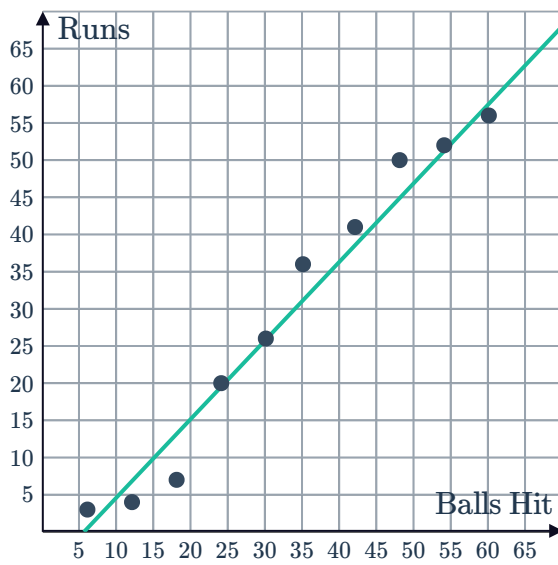
i Approx. 91

ii Approx. 152

c Increases

21

a



b

i Approx. 23

ii Approx. 63

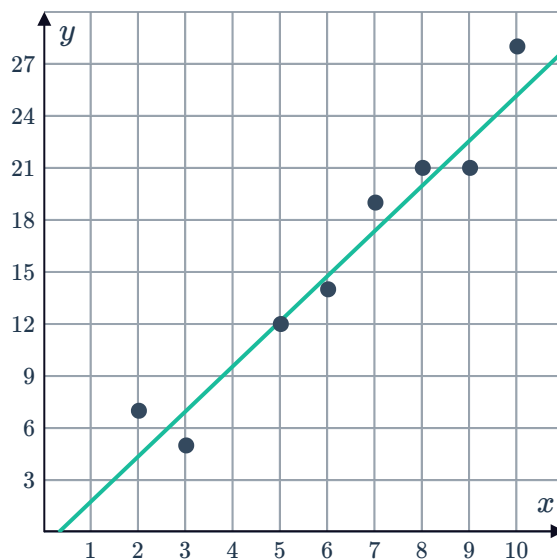
c Positive

22

a Positive



b



c D

23

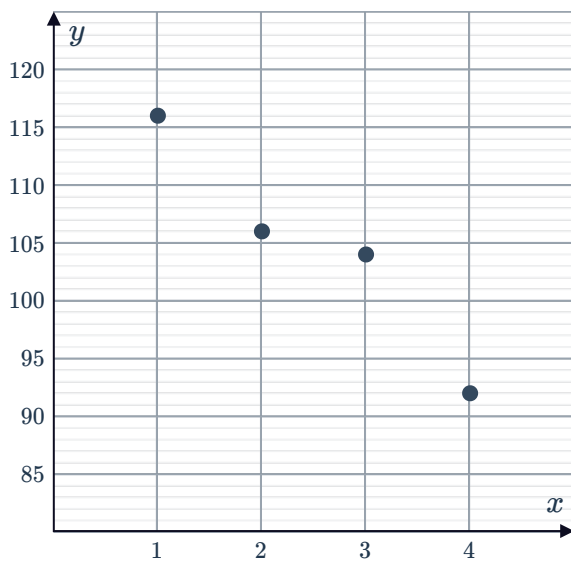
a Negative

b C

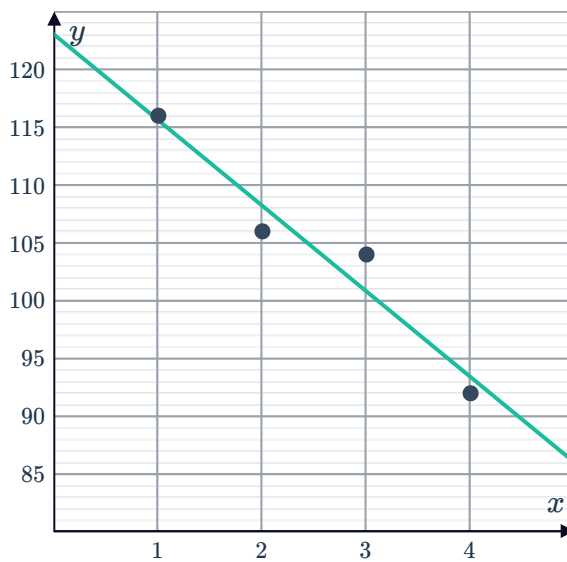
24 4 knots

25

a

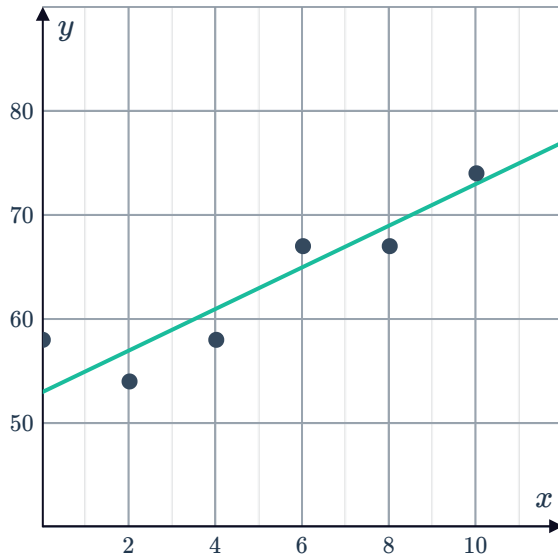


b



26

a



b (10, 74)

c (0, 58)

27

a 20 years

b 1300 fish

c 11 years

28

a 3 seconds

b 2.5 hours

29

a 20°C

b 210 chirps per minute

c 130 chirps per minute

30

a 8500 m

b 6500 m

c 300 seconds

d 1800 seconds

31

a 90. When a student watched no TV, their average result in the exam was 90.

b Yes. When a student watches 0 hours, their time is within the data range, and the result makes sense.

c No

32

a 30

b When a child doesn't read any pages each day, then their vocabulary, on average, is 30 words.

c Yes. When a child reads 0 pages a day, the number of pages is within the data range and the value of the vocabulary makes sense.

d No

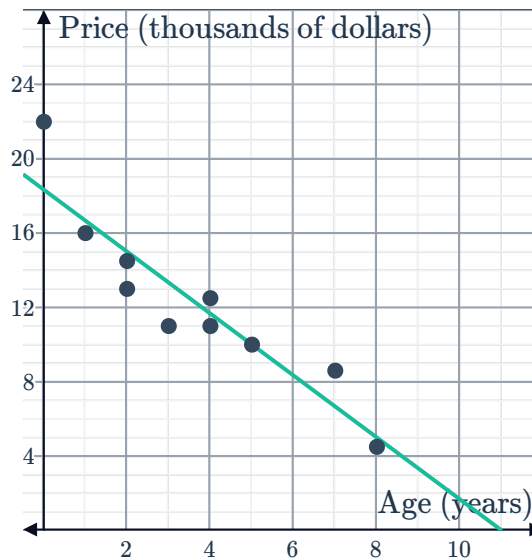


33

- a 100
- b When a student spends no time using the app, they achieve a math score of 100.
- c Yes. When a student spends 0 minutes using the app, their time is within the data range and the value of the score makes sense.
- d No

34

a



- b (0, 18 000)
- c A Toyota Camry that is 0 years old costs \$18 000.
- d 11 years

35

- a 14 811
- b The y -intercept indicates that when a house is brand new, its value is, on average, \$14 811.
- c No. When a house is zero years old its age is not within the data range.
- d Approximately 4.9 years.
- e \$1 200 000